

HW 3. $\Delta W_j \sim \sqrt{h} Z_j$

$$P1. E\left[\sum_{j=0}^{n-1} (\Delta W_j)^4\right] = \sum_{j=0}^{n-1} E[(\Delta W_j)^4] \\ = \sum_{j=0}^{n-1} E[h^2 Z_j^4] = h^2 \sum_{j=0}^{n-1} E[Z_j^4] \\ = h^2 \cdot n \cdot 3 = h \cdot T \cdot 3$$

 $\rightarrow 0$ as $h \rightarrow 0$

$$\text{Var}\left(\sum_{j=0}^{n-1} (\Delta W_j)^4\right) = \text{Var}\left(\sum_{j=0}^{n-1} h^2 Z_j^4\right) \\ = h^4 n \text{Var}(Z^4) \\ = h^4 n (E[Z^8] - (E[Z^4])^2) \\ = h^3 T \cdot (105 - 9)$$

 $\rightarrow 0$ as $h \rightarrow 0$

$$\text{Thus } \sum_{j=0}^{n-1} (\Delta W_j)^4 \xrightarrow{\text{a.s.}} 0, \text{ as } h \rightarrow 0.$$

$$P2. f(t, W_t) = 2t + t + e^{W_t} = X_t.$$

$$f_t = 1, f_W = e^{W_t}, f_{WW} = e^{W_t}$$

$$\Rightarrow dX_t = (f_t + \frac{1}{2} f_{WW}) dt + f_W dW_t \\ = (1 + \frac{1}{2} e^{W_t}) dt + e^{W_t} dW_t \\ = \frac{1}{2} (X_t - t) dt + (X_t - t - 2) dW_t$$

$$P3. X_t = f(t, W_t) = \cos(t^2 + W_t)$$

$$f_t = -2t \sin(t^2 + W_t)$$

$$f_W = -\sin(t^2 + W_t)$$

$$f_{WW} = -\cos(t^2 + W_t)$$

$$\Rightarrow dX_t = [-2t \sin(t^2 + W_t) - \frac{1}{2} \cos(t^2 + W_t)] dt \\ - \sin(t^2 + W_t) dW_t \\ = [-2t \sqrt{1 - X_t^2} - \frac{1}{2} X_t] dt \\ - \sqrt{1 - X_t^2} dW_t.$$

P4. Straightforward.

$$P5. dI_t^{(1)} = dW_t = I_t^{(0)} dW_t \quad \text{since } I_t^{(0)} = 1$$

$$dI_t^{(2)} = d(W_t^2 - t) \\ = 0 \cdot dt + 2W_t dW_t = 2I_t^{(1)} dW_t.$$

$$dI_t^{(3)} = d(W_t^3 - 3tW_t) \\ = 0 \cdot dt + (3W_t^2 - 3t) dW_t = 3I_t^{(2)} dW_t$$

$$dI_t^{(4)} = d(W_t^4 - 6tW_t^2 + 3t^2) \\ = 0 \cdot dt + (4W_t^3 - 12tW_t) dW_t \\ = 4I_t^{(3)} dW_t.$$

$$dI_t^{(5)} = 5I_t^{(4)} dW_t \\ = (5W_t^4 - 30tW_t^2 + 15t^2) dW_t.$$

$$\text{Assume } I_t^{(5)} = f(t, W_t)$$

$$\Rightarrow \begin{cases} f_W = 5W_t^4 - 30tW_t^2 + 15t^2 - \phi \\ f_t + \frac{1}{2} f_{WW} = 0 \end{cases} \quad \text{--- ③}$$

$$\phi \Rightarrow f(t, W_t) = W_t^5 - 10tW_t^3 + 15t^2W_t + h(t) \quad \text{--- ④}$$

$$\text{put ③ into ④} \Rightarrow \\ -10W_t^3 + 30tW_t + h'(t) + \frac{1}{2} (20W_t^3 - 60tW_t^2) = 0 \\ \Rightarrow h'(t) = 0 \\ \Rightarrow h(t) = C$$

We simply take $h(t) = 0$.

$$\Rightarrow I_t^{(5)} = W_t^5 - 10tW_t^3 + 15t^2W_t$$

Similarly, we can get

$$I_t^{(6)} = W_t^6 - 15tW_t^4 + 45t^2W_t^2 - 15t^3$$

$$P6. (a). dY_t = d \sinh X_t \rightarrow X_t = dt + dW_t$$

$$= d \frac{e^{X_t} - e^{-X_t}}{2} \\ = \left(1 \cdot \frac{e^{X_t} + e^{-X_t}}{2} + \frac{1}{2} \cdot \frac{e^{X_t} - e^{-X_t}}{2} \right) dt \\ + \frac{e^{X_t} + e^{-X_t}}{2} dW_t$$

$$= \left(\frac{1}{2} Y_t + \cosh X_t \right) dt + \cosh X_t dW_t \\ = \left(\frac{1}{2} Y_t + \sqrt{1 + Y_t^2} \right) dt + \sqrt{1 + Y_t^2} dW_t$$

$$(b) Z_t = Y_t^3 \rightarrow \text{let } \mu = \frac{1}{2} Y_t + \sqrt{1+Y_t^2}, \sigma = \sqrt{1+Y_t^2}$$

$$dZ_t = dY_t^3$$

$$= (\mu \cdot 3Y_t^2 + \frac{1}{2}\sigma^2 \cdot 6Y_t)dt + \sigma \cdot 3Y_t^2 dW_t$$

$$= (3Y_t + \frac{9}{2}Y_t + 3\sqrt{1+Y_t^2}Y_t^2)dt + 3\sqrt{1+Y_t^2}Y_t^2 dW_t$$

$$= (3\sqrt[3]{Z_t} + \frac{9}{2}Z_t + 3\sqrt{1+Z_t^{2/3}}Z_t^{2/3})dt + 3\sqrt{1+Z_t^{2/3}}Z_t^{2/3}dW_t$$